

Chapter 3 - Rings and Moduls of Fractions

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22 Feb 2026

Exercise 1

(i) ($\Rightarrow$): Let M be generated by $\{m_1, m_2, \dots, m_n\}$, for $\forall i \in [1, n] \cap \mathbb{Z}$, we have

$$\forall r, t \in S, \exists s \in S, \text{s.t. } m_i s(r - t) = 0, \text{i.e. } m_i s = 0$$

Denote the s above as s_i . Since M is finitely-generated, we can let

$$s = s_1 s_2 \dots s_n$$

And have $sM = 0$.

(ii) ($\Leftarrow$): When such s exist, then for $\forall \frac{m_1}{s_1}, \frac{m_2}{s_2} \in S^{-1}M$,

$$s(s_1 m_2 - s_2 m_1) = 0 \implies \frac{m_1}{s_1} = \frac{m_2}{s_2}$$

So $S^{-1}M = 0$.

□

Exercise 2

By *Proposition 1.9*, the problem is equivalent to prove that for any $\frac{a}{s} \in S^{-1}A$, we have

$$\forall \frac{b}{t} \in S^{-1}A, 1 - \frac{a}{s} \frac{b}{t} \text{ is a unit in } S^{-1}A$$

Since $1 + \mathfrak{a}$ is a multiplicative set, $\exists c \in \mathfrak{a}, \text{s.t. } st = 1 + c$. Now

$$\begin{aligned} 1 - \frac{a}{s} \frac{b}{t} &= 1 - \frac{ab}{st} \\ &= 1 - \frac{ab}{1+c} \\ &= \frac{1+c-ab}{1+c} \end{aligned}$$

And we can easily see $\frac{1+c}{1+c-ab} = 1 + \frac{ab}{1+c-ab} \in S^{-1}A$. So the inverse of $1 - \frac{a}{s} \frac{b}{t}$ is $1 + \frac{ab}{1+c-ab}$.

□

Exercise 3

Define

$$f : (ST)^{-1}A \longrightarrow U^{-1}(S^{-1}A)$$

by mapping

$$\frac{a}{st} \longmapsto \frac{\frac{a}{s}}{\frac{t}{1}}$$

We'll prove f is an isomorphism:

(i) Surjection: Trivial.

(ii) Injection:

$$\frac{\frac{a_1}{s_1}}{\frac{t_1}{1}} = \frac{\frac{a_2}{s_2}}{\frac{t_2}{1}}$$

$\Leftrightarrow$

$$\exists t \in T, \text{ s.t. } \frac{t}{1} \left(\frac{a_1 t_2}{s_1} - \frac{a_2 t_1}{s_2} \right) = 0$$

$\Leftrightarrow$

$$\exists t \in T, \text{ s.t. } t(a_1 t_2 s_2 - a_2 t_1 s_1) = 0$$

$\Leftrightarrow$

$$\frac{a_1}{s_1 t_1} = \frac{a_2}{s_2 t_2}$$

So f is injective.

(iii) Homomorphism: Trivial.

□

Exercise 4

Define

$$f : S^{-1}B \longrightarrow T^{-1}B$$

by

$$\frac{b}{s} \longmapsto \frac{b}{f(s)}$$

We'll prove it's an isomorphism:

(i) Homomorphism: Trivial

(ii) Surjection: Trivial

(iii) Injection:

$$\frac{b_1}{f(s_1)} = \frac{b_2}{f(s_2)}$$

$\Leftrightarrow$

$$\exists s \in S, \text{ s.t. } f(s) (b_1 f(s_2) - b_2 f(s_1)) = 0$$

$\Leftrightarrow$

$$\frac{b_1}{s_1} = \frac{b_2}{s_2}$$

□

Exercise 5

(i) By *Proposition 3.11-(v)*, the operation S^{-1} commutes with the formations of radicals. So $\forall \mathfrak{p} \subseteq A$,

$$(\text{Nil}(A))_{\mathfrak{p}} = \text{Nil}(A_{\mathfrak{p}}) = 0$$

Therefore by *Proposition 3.8*, $\text{Nil}(A_{\mathfrak{p}}) = 0$.

(ii) No. Counterexample given here

Exercise 6

(i) Σ is nonempty, therefore has maximal elements by *Zorn's Lemma*.

- (ii) (I) When $S \in \Sigma$ is maximal, we'll prove $A \setminus S$ is a minimal prime ideal. $\forall x, y \in A \setminus S_0$, let

$$S_x = \{sx^n \mid s \in S, n \in \mathbb{Z}^+ \cup \{0\}\}, S_y = \{sy^n \mid s \in S, n \in \mathbb{Z}^+ \cup \{0\}\}$$

Since $S \in \Sigma$ is maximal and $S_x, S_y \supsetneq S$, we have $S_x, S_y \notin \Sigma, 0 \in S_x, 0 \in S_y$. Therefore $\exists s_1, s_2 \in S, n_1, n_2 \in \mathbb{Z}^+$, such that

$$s_1 x^{n_1} = s_2 x^{n_2} = 0$$

And we can see

$$s_1 s_2 (x + y)^{n_1 + n_2 - 1} = 0, s_1 s_2 (xy)^{n_1 n_2} = 0$$

Which is to say $x + y, xy \in A \setminus S$, and we can easily obtain $A \setminus S$ is an ideal. The prime property is trivial, and the maximal property can be obtained directly from (II) and the maximal property of S .

- (II) When $A \setminus S$ is a minimal prime ideal of A : Denote $I = A \setminus S$. For $\forall x, y \in S$, we have $x, y \notin I$. If $xy \in I$, by the prime property of I we will immediately obtain $x \in I$ or $y \in I$, which is a contradictory. So $xy \notin I, xy \in S$, S is a multiplicatively closed subset. Also, if S is not maximal in Σ , the one in Σ strictly contains S will correspond to a smaller prime ideal in $A \setminus S$, which contradicts with the minimal property of I .

□

Exercise 7

- (i) (I) ($\Leftarrow$): When $A \setminus S$ is a union of prime ideals, $x \in A \setminus S$ or $y \in A \setminus S$ will immediately implies $xy \notin S$. So $xy \in S$ will imply $x, y \in S$.

- (II) ($\Rightarrow$): We'll prove that any elements in $A \setminus S$ was contained in a prime ideal which is disjoint to S : $\forall a \in A \setminus S, (a) \cap S \subseteq \{0\}$, otherwise if $a^n = a \cdot a^{n-1} \in S$, we'll have $a \in S$, contradictory.

Now we consider Σ be the set of all ideals containing (a) and being disjoint to S . By *Zorn's Lemma*, we can pick out a maximal element, namely $\mathfrak{p} \subseteq \Sigma$. We'll prove $\mathfrak{p}$ is prime:

Suppose we have $xy \in \mathfrak{p}, x, y \notin \mathfrak{p}$. Since $\mathfrak{p}$ is maximal and $\mathfrak{p} + (x), \mathfrak{p} + (y) \supsetneq \mathfrak{p}$, we know both of $\mathfrak{p} + (x), \mathfrak{p} + (y)$ intersects with S . Therefore $\exists s_1, s_2 \in S$ such that $\exists p_1, p_2 \in \mathfrak{p}, a_1, a_2 \in A$,

$$s_1 = p_1 + a_1 x, s_2 = p_2 + a_2 x$$

Then we have

$$(s_1 - p_1)s_2 \in S$$

whereas

$$\begin{aligned} (s_1 - p_1)s_2 &= a_1 x(p_2 + a_2 y) \\ &= a_1 p_2 x + a_1 a_2 xy \in \mathfrak{p} \end{aligned}$$

Which means $(s_1 - p_1)s_2 \in S \cap \mathfrak{p}$, contradicting with the construction of $\mathfrak{p}$.

□

- (ii) By (i), $\overline{S}$ defined by the complement in A of union of prime ideals disjoint from S is indeed a saturated multiplicative closed subset of A .

We only have to prove $\overline{S}$ is the unique minimal one, which is trivial since the family of sets, which consists of prime ideals disjoint from S , has a unique maximal element (final object, where morphisms taken under inclusion).

- (iii)

$$\mathfrak{p} \cap S \subsetneq \{0\}$$

$\Leftrightarrow$

$$\exists p \in \mathfrak{p}, a \in \mathfrak{a}, \text{ s.t. } 1 + p = a$$

$\Leftrightarrow$

$$\mathfrak{a} + \mathfrak{p} = (1)$$

$\Leftrightarrow$

$\mathfrak{a}, \mathfrak{p}$ are coprime

Conversely, it's easy to prove that $\mathfrak{a}, \mathfrak{p}$ being coprime will implies $\mathfrak{p} \cap S \subseteq \{0\}$. Therefore $\overline{S}$ is the complement of the union of all prime ideals which are not coprime with $\mathfrak{a}$.

Exercise 8

(I) $(i) \Rightarrow (ii)$: $\forall t \in T, \phi(\frac{t}{1}) = \frac{t}{1}$ is a unit in $T^{-1}A$. Since ϕ is bijective, the preimage of the inverse of $\frac{t}{1} \in T^{-1}A$ is also the inverse of $\frac{t}{1} \in S^{-1}A$.

(II) $(ii) \Rightarrow (iii)$: $\forall t \in T, \frac{t}{1}$ is a unit in $S^{-1}A$ and we denote $\frac{a}{s} = \left(\frac{t}{1}\right)^{-1}$. Therefore $\exists s_0 \in S$, s.t. $s_0(at-s) = 0$, which indicate

$$(s_0a)t = s_0s \in S$$

Thus we can let $x = s_0a$.

(III) $(iii) \Rightarrow (iv)$: Suppose $\exists t \in T$, s.t. t is contained in a prime ideal $\mathfrak{p}$ which is disjoint from S . By (iii) , $\exists x \in A$, s.t. $xt \in S$. However, $t \in \mathfrak{p} \Rightarrow xt \in \mathfrak{p} \Rightarrow xt \in S \cap \mathfrak{p}$, contradictory.

(IV) $(iv) \Rightarrow (v)$: Otherwise T cannot be contained in $\overline{S}$.

(V) $(v) \Rightarrow (ii)$: Suppose $\frac{t}{1}$ is not a unit in $S^{-1}A$. Then there exist a maximal ideal $\mathfrak{m} \subseteq S^{-1}A$, s.t. $\frac{t}{1} \in \mathfrak{m}$.

Consider the natural homomorphism $f : A \rightarrow S^{-1}A$ and denote $f^{-1}(\mathfrak{m}) = \mathfrak{p}$. It immediately follows that $\mathfrak{p} \cap S = \emptyset$ (otherwise $f(\mathfrak{p}) = S^{-1}A$). Also, from $\frac{t}{1} \in \mathfrak{m}$ we have $t \in \mathfrak{p}$, i.e. $\mathfrak{p}$ meets T , therefore $\mathfrak{p}$ meets S , contradictory.

(VI) $(ii) \Rightarrow (i)$: Elements in $T^{-1}A$ generally has the form $\frac{a}{t}$. Since $\frac{t}{1}$ is a unit in $S^{-1}A$, we have

$$a \cdot \left(\frac{t}{1}\right)^{-1} \in S^{-1}A, \phi\left(a \cdot \left(\frac{t}{1}\right)^{-1}\right) = \frac{a}{t}$$

hence ϕ is surjective.

On the other hand, suppose $\phi\left(\frac{a}{s}\right) = 0$ where $\frac{a}{s} \neq 0$. Then $\exists t \in T \setminus S, at = 0$. In this case $\frac{t}{1}$ cannot be unit in $S^{-1}A$.

□

Exercise 9

(I) We follow from the hint: Denote the family of all multiplicatively closed set which doesn't contains 0 by Σ , the problem is equivalent to prove that all the maximal elements in Σ must contains S .

We prove by contradictory and suppose there exist a maximal element $S \in \Sigma$ such that $\exists a \in S_0 \setminus S$. Let $S_a = \{sa^n \mid s \in S, n \in \mathbb{Z}^+ \cup \{0\}\}$, which is a multiplicatively closed set strictly contains S . Because S is maximal in Σ , $S_a \notin \Sigma, 0 \in S_a$. Hence, $\exists n \in \mathbb{Z}^+$, s.t.

$$sa^n = 0$$

However, from $a \in S_0 \setminus S$ we know a^n is a non-zero-divisor, i.e. $s = 0$, which is absurd. Hence the proof was done.

□

(II) (i) Homomorphism $A \rightarrow S_0^{-1}A$ is injective iff $\forall s \in S_0, a \in A \setminus \{0\}, sa \neq 0$. This is to say S_0 consists of non-zero-divisors.

(ii) Every element in $A \setminus S_0$ will be mapped to a zero-divisor while every element in S_0 will be mapped to a unit, obviously.

(iii) Injectivity is obtained from (i), and the preimage of $\frac{a}{s} \in S^{-1}A$ is $as^{-1} \in A$, obviously.

□

Remark 1. From **Exercise 3.6, 3.9**, we can see saturated sets and sets made up of prime-ideals-union acts as complements to each other. And if we let S be maximal, the corresponding $A \setminus S$ will be left to be only one prime idea.

Exercise 10

(ii) Hinited

- (i) By *Proposition 2.19(iv)*, for any $S^{-1}A$ -module N , if we have $S^{-1}A$ -module M, M' and homomorphism $f : M \rightarrow M'$ such that

$$0 \rightarrow M \xrightarrow{f} M'$$

exact as $S^{-1}A$ -module sequence, we only need to prove

$$0 \rightarrow S^{-1}M \otimes_{S^{-1}A} S^{-1}N \xrightarrow{f \otimes 1} S^{-1}M' \otimes_{S^{-1}A} S^{-1}N$$

is exact.

Notice that restriction on the rings under which the modules are only affect the structure of scalar multiplication while holding the image and kernel, we may view N, M, M' as A -module and f as A -module homomorphism while having

$$0 \rightarrow M' \xrightarrow{f|_A} M$$

exact as A -module sequence.

Therefore from the absolute flatness of A we may have N as a flat A -module, i.e.

$$0 \rightarrow M' \otimes_A N \xrightarrow{f|_A \otimes 1} M \otimes_A N$$

is exact as a A -module sequence.

Now by *Proposition 3.3*, the S^{-1} -functor is exact, and

$$0 \rightarrow S^{-1}(M' \otimes_A N) \xrightarrow{S^{-1}(f|_A \otimes 1)} S^{-1}(M \otimes_A N)$$

is an exact $S^{-1}A$ sequence. Moreover, by *Proposition 3.7* such sequence is isomorphic to

$$0 \rightarrow S^{-1}M \otimes_{S^{-1}A} S^{-1}N \xrightarrow{f \otimes 1} S^{-1}M \otimes_{S^{-1}A} S^{-1}N$$

and we get its exactness.

- (ii) (a) When A is absolutely flat: From (i) $A_{\mathfrak{m}}$ is flat for any maximal ideal $\mathfrak{m} \subseteq A$. And by *Exercise 2.28*, a flat local ring must be a field, i.e. $A_{\mathfrak{m}}$ is a field.
- (b) When for any maximal ideal $\mathfrak{m} \subseteq A$, $A_{\mathfrak{m}}$ is a field: For any A -module N , consider $N_{\mathfrak{m}}$ as a $A_{\mathfrak{m}}$ -module. Since $A_{\mathfrak{m}}$ is a field, $N_{\mathfrak{m}}$ is a vector space, therefore has a basis and is a free module over $A_{\mathfrak{m}}$. It's well-known that a free module must be flat, therefore $N_{\mathfrak{m}}$ is flat over $A_{\mathfrak{m}}$. Because $\mathfrak{m}$ is arbitrarily chosen, by *Proposition 3.10* we know N is flat, and therefore A is absolutely flat.

□

Exercise 11

(UNFINISHED)

- (I) (a) (i) $\Leftrightarrow$ (ii):

$$(i) \Leftrightarrow$$

$$(A/\text{Nil } A)_{\mathfrak{m}} \text{ is a field for all maximal ideals } \mathfrak{m} \text{ of } A/\text{Nil } A$$

$$\Leftrightarrow$$

$$(A/\text{Nil } A)_{\mathfrak{m}} \text{ only has ideals } (1), (0) \text{ is a field for all maximal ideals } \mathfrak{m} \text{ of } A/\text{Nil } A$$

$$\Leftrightarrow$$

$$(A/\text{Nil } A)_{\mathfrak{m}} \text{ only has prime ideal } (0) \text{ for all maximal ideals } \mathfrak{m} \text{ of } A/\text{Nil } A$$

$\Leftrightarrow$

$A/\text{Nil } A$ has an unique prime ideal contained in $\mathfrak{m}$ for all maximal ideals $\mathfrak{m}$ of $A/\text{Nil } A$

$\Leftrightarrow$

Every prime ideals of $A/\text{Nil } A$ is maximal

$\Leftrightarrow (ii)$

- (b) $(ii) \Rightarrow (iii)$: When all prime ideals are maximal ideals, then every subsets, which consists of single point, are the final objects in the category of $\text{Spec } A$ (morphisms under inclusion), therefore closed.
- (c) $(iii) \Rightarrow (iv)$:
- (d) $(iv) \Rightarrow (i)$:

(II)

Exercise 12

It's trivial that torsion elements of M form a submodule of M .

- (i) Suppose $\exists x \notin T(M)$, s.t. $\exists a \in A, ax = 0$, i.e. $ax \in T(M)$. Then $\exists b \in A$, s.t. $b(ax) = 0$, therefore $abx = 0, x \in T(M)$, contradictory.
- (ii) $\forall m \in T(M), \exists a \in A$, s.t. $am = 0 \Rightarrow af(m) = 0, f(m) \in T(N)$.
- (iii) *(My proof might be a bit ambiguous, but I think I did proved it out.)*

Denote

$$0 \xrightarrow{f} M' \xrightarrow{g} M \xrightarrow{h} X''$$

and we need to prove

$$0 \xrightarrow{\bar{f}} T(M') \xrightarrow{\bar{g}} T(M) \xrightarrow{\bar{h}} T(X'')$$

is exact.

First, $\text{Ker } \bar{h} = \text{Ker } h \cap T(M'') = \text{Im } g \cap T(M'')$. To prove $\text{Ker } \bar{h} = \text{Im } \bar{g}$, we need to show

$$g(T(M')) = g(M') \cap T(M)$$

i.e. $g(M' \setminus T(M')) \cap T(M) \subseteq \{0\}$.

Suppose $\exists m' \in M' \setminus T(M'), g(m') \in T(M)$. Then $\exists a \in A, ag(m') = 0 \Rightarrow g(am') = 0 \Rightarrow am' \in \text{Ker } g$
Which is to say

$$am' \in \text{Ker } g = \text{Im } f = \{0\}$$

Contradictory. Hence we have $\text{Ker } \bar{h} = \text{Im } \bar{g}$.

Also, the case $\text{Ker } \bar{g} = \text{Im } \bar{f}$ is trivial.

- (iv) *(The hint seems to be complicated, I'll just follow my own thought.)*

Denote

$$\begin{aligned} \varphi : M &\longrightarrow K \otimes_A M \\ x &\longmapsto 1 \otimes x \end{aligned}$$

First, for all $m \in T(M)$, suppose $am = 0$. Then

$$\varphi(m) = 1 \otimes m = \frac{1}{a} \otimes am = 0$$

So $T(M) \subseteq \text{Ker } \varphi$.

On the other hand, $\forall m \in \text{Ker } \varphi, \varphi(m) = 1 \otimes m = 0$. A is an integral domain, so there're no zero divisors in K . Thus $1 \otimes m$ must be annihilated in the m term, i.e. $\exists a \in A, am = 0 \Rightarrow m \in T(M)$.

□

Exercise 13

(I) On one hand, $\forall \frac{m}{s} \in S^{-1}M, \exists a \in A, \text{s.t. } a \cdot \frac{m}{s} = 0 \Rightarrow \frac{am}{s} = 0, m \in T(M), \frac{m}{s} \in S^{-1}(T(M))$. So $S^{-1}M \subseteq S^{-1}(T(M))$.

On the other hand, for all $\frac{m}{s} \in S^{-1}(T(M))$: By $m \in T(M)$ we have $a \in A, \text{s.t. } ma = 0$. So $a \cdot \frac{m}{s} = 0, \frac{m}{s} \in S^{-1}M$.

(II) By (I),

$$T(M_{\mathfrak{p}}) = (A \setminus \mathfrak{p})^{-1}T(M) = (T(M))_{\mathfrak{p}}$$

$$T(M_{\mathfrak{m}}) = (A \setminus \mathfrak{m})^{-1}T(M) = (T(M))_{\mathfrak{m}}$$

Use *Proposition 3.8* and we're done. □

Exercise 14

We follow the hint: By *Exercise 2.2*, $(A/\mathfrak{a}) \otimes_A M \cong M/\mathfrak{a}M$. So we have $M_{\overline{\mathfrak{m}}} = 0$ for all maximal ideals $\overline{\mathfrak{m}}$ in $A/\mathfrak{a}$, which correspond to $\mathfrak{m}$ in A . Therefore by *Proposition 3.8*, we have $M/\mathfrak{a}M = 0$, i.e. $M = \mathfrak{a}M$. □

Exercise 15

(UNFINISHED)

(Following the hint given, the idea is that we want to use the theories in vector spaces to solve this problem, so we would try to turn the original modules into vector spaces.)

Following the notations in the given hint, for exact sequence

$$0 \longrightarrow N \longrightarrow F \longrightarrow F \longrightarrow 0$$

we only need to prove

$$0 \longrightarrow k \otimes_A N \longrightarrow k \otimes_A F \longrightarrow k \otimes_A F \longrightarrow 0$$

is exact, which we only have to prove $k = A/\mathfrak{m}$ is exact.

Exercise 16

(a) (i) $\Rightarrow$ (ii): Denote the homomorphism $A \longrightarrow B$ by φ . $\forall \mathfrak{p} \in \text{Spec } A, \mathfrak{p}^{ec} = \mathfrak{p}$. So by *Proposition 3.16*, $\mathfrak{p}$ is the contraction of a prime ideal of B .

(b) (ii) $\Rightarrow$ (iii): For any maximal ideal $\mathfrak{m}$ there exist a prime ideal of B , namely $\mathfrak{p}$, such that $\mathfrak{m} = \mathfrak{p}^c$. So $\mathfrak{m}^e = B\varphi(\mathfrak{m}) = \mathfrak{p} \subsetneq (1)$.

(c) ()